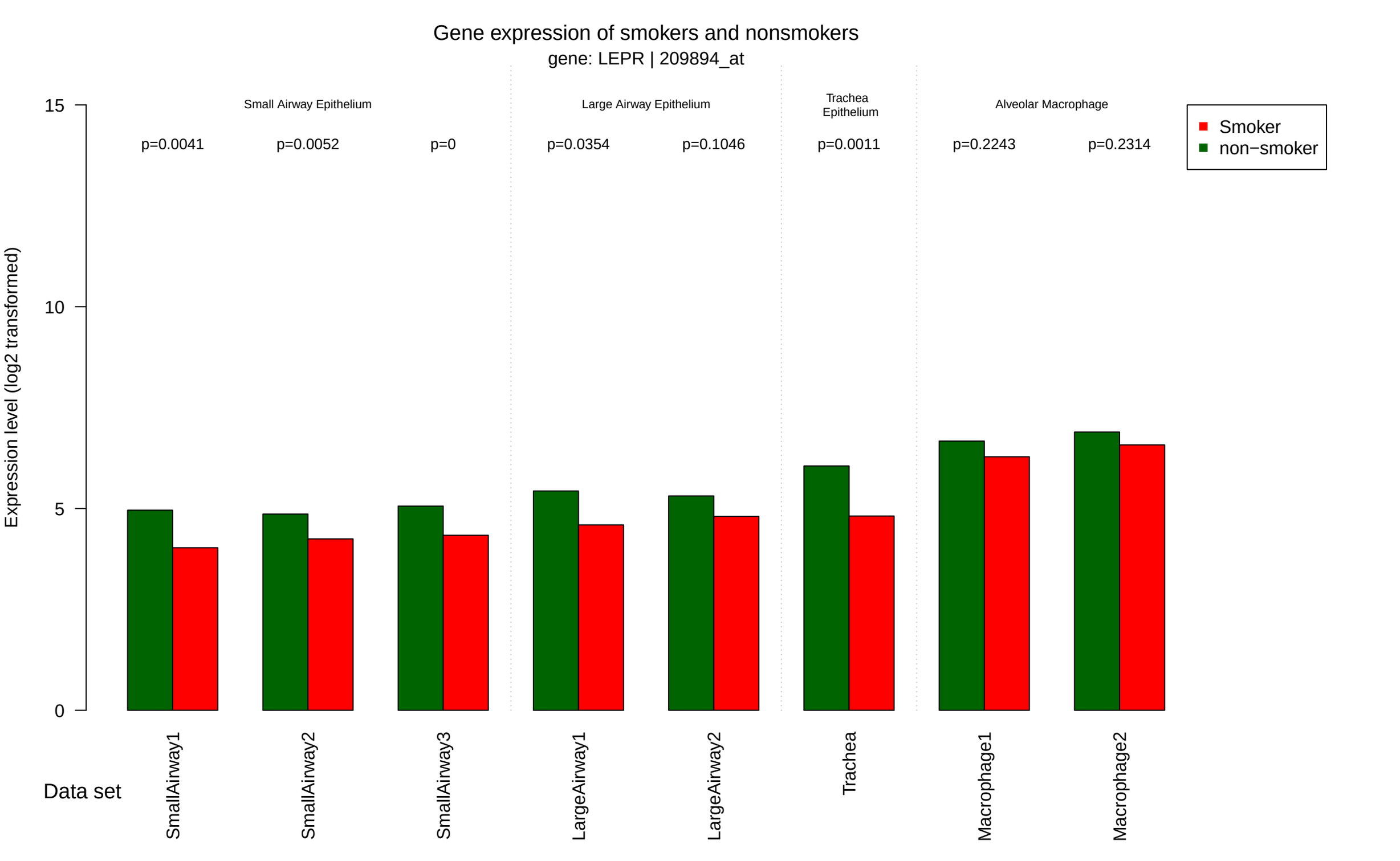
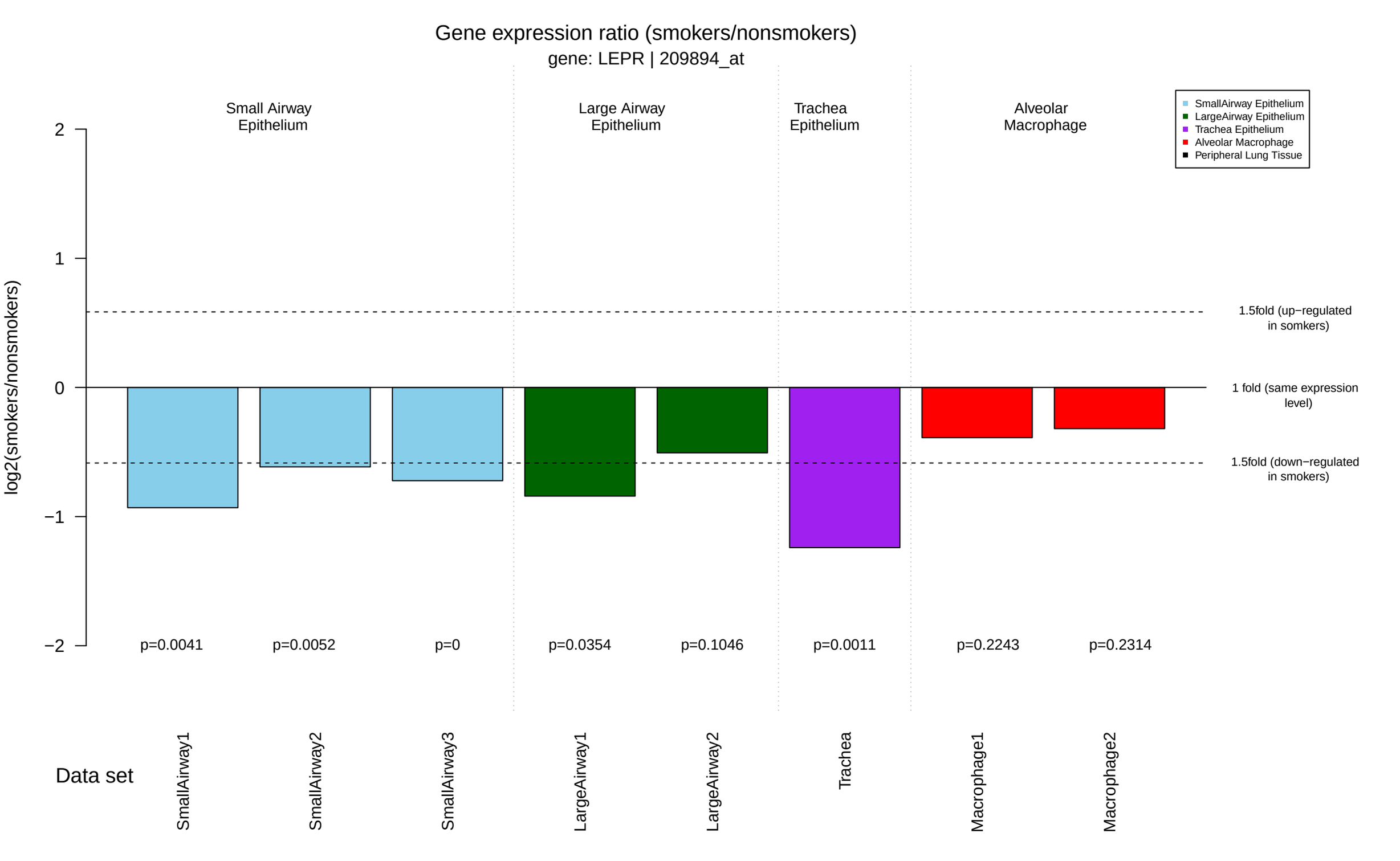
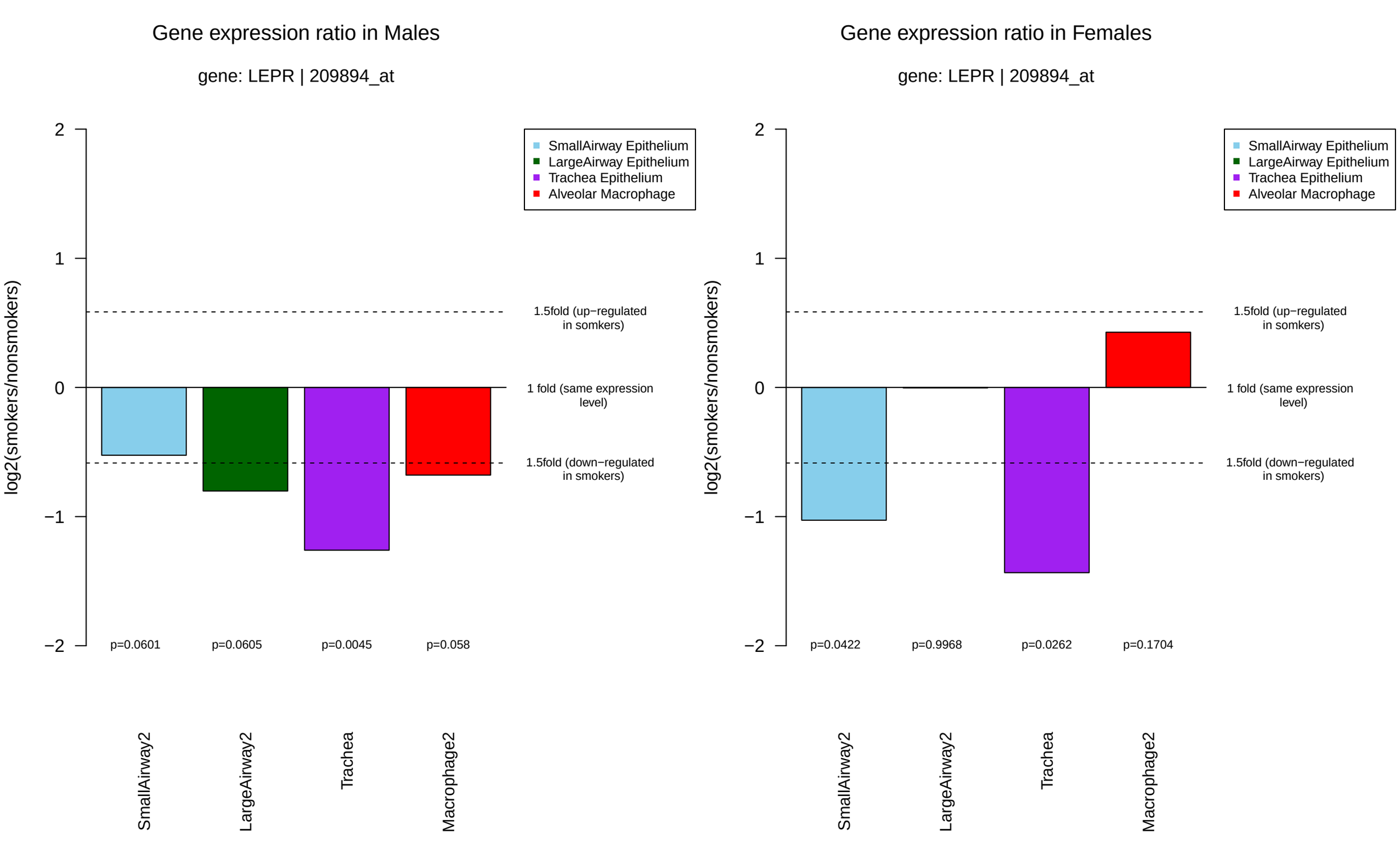
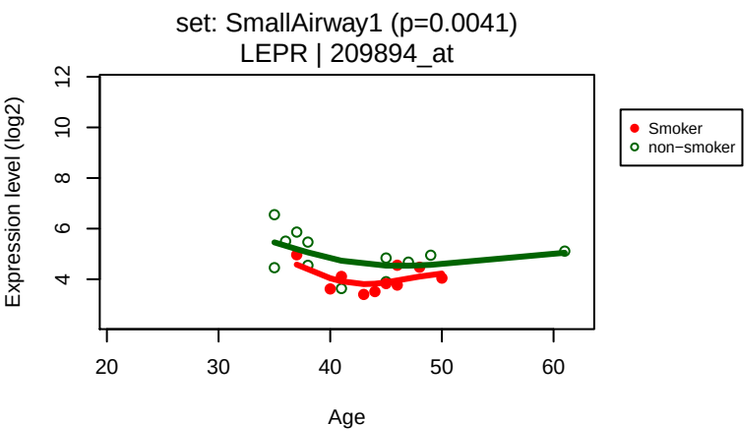
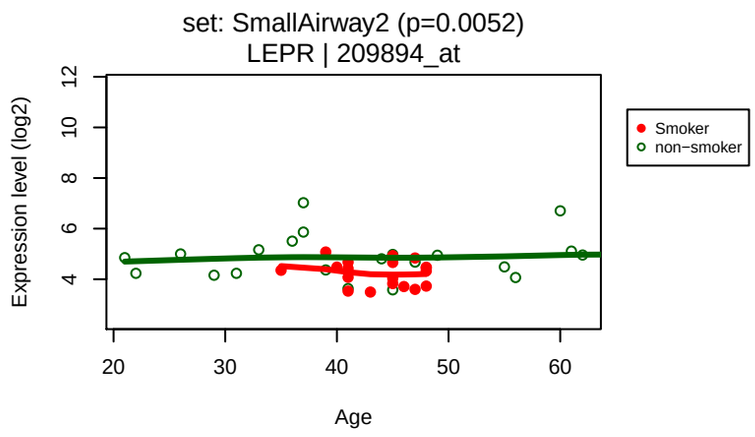
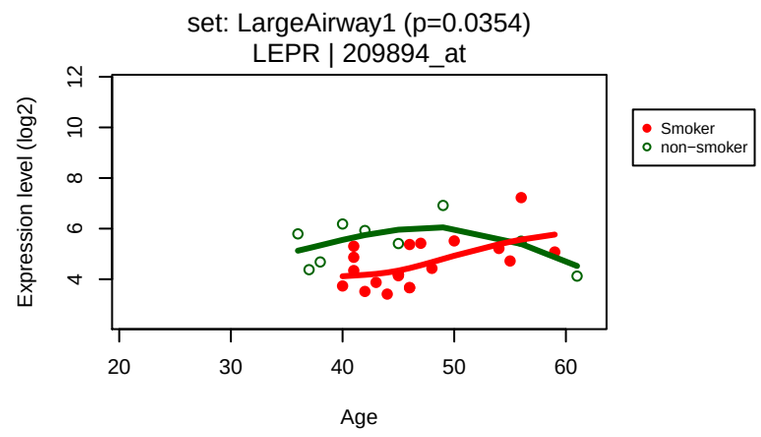

Gene expression of smokers and nonsmokers
gene: LEPR | 209894_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0041
p=0.0052
p=0
p=0.0354
p=0.1046
p=0.0011
p=0.2243
p=0.2314
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

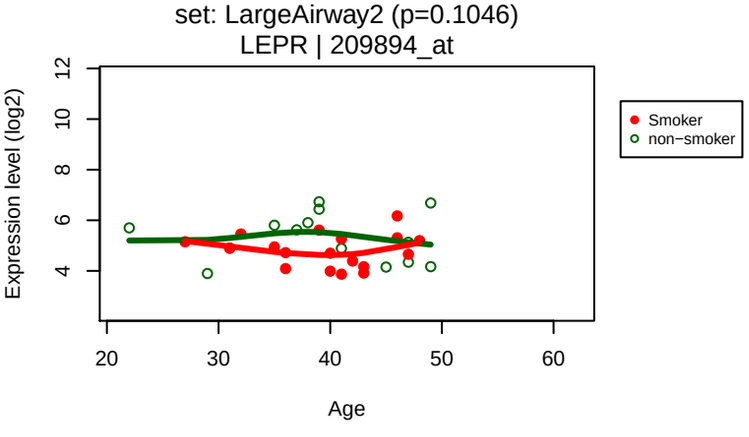

Gene expression ratio (smokers/nonsmokers)
gene: LEPR | 209894_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0041
p=0.0052
p=0
p=0.0354
p=0.1046
p=0.0011
p=0.2243
p=0.2314
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

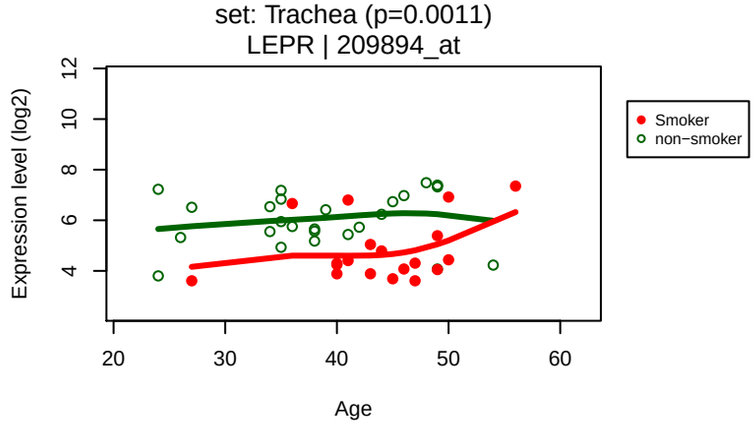

Gene expression ratio in Males
gene: LEPR | 209894_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0601
p=0.0605
p=0.0045
p=0.058
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LEPR | 209894_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0422
p=0.9968
p=0.0262
p=0.1704
SmallAirway2
LargeAirway2
Trachea
Macrophage2

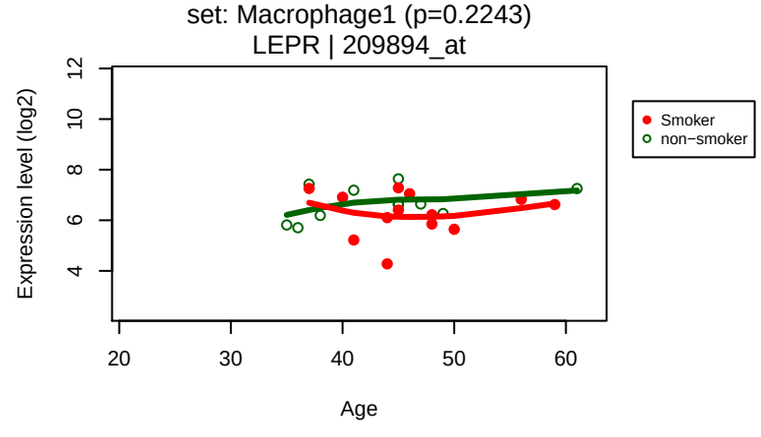

set: SmallAirway1 (p=0.0041)
LEPR | 209894_at
Expression level (log2)
Age
Smoker
non−smoker

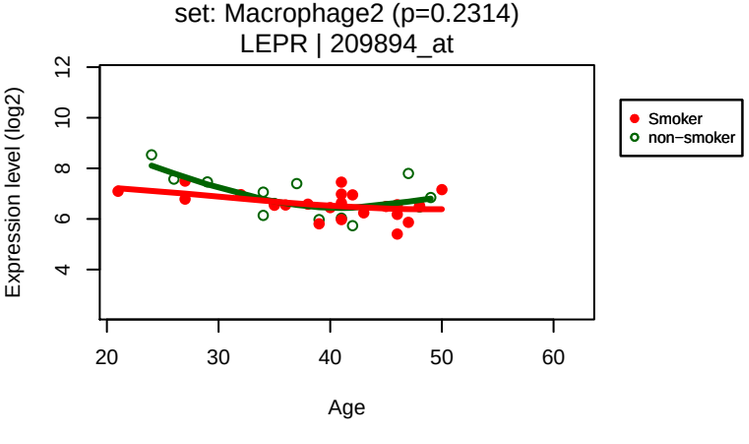

set: SmallAirway2 (p=0.0052)
LEPR | 209894_at
Expression level (log2)
Age
Smoker
non−smoker

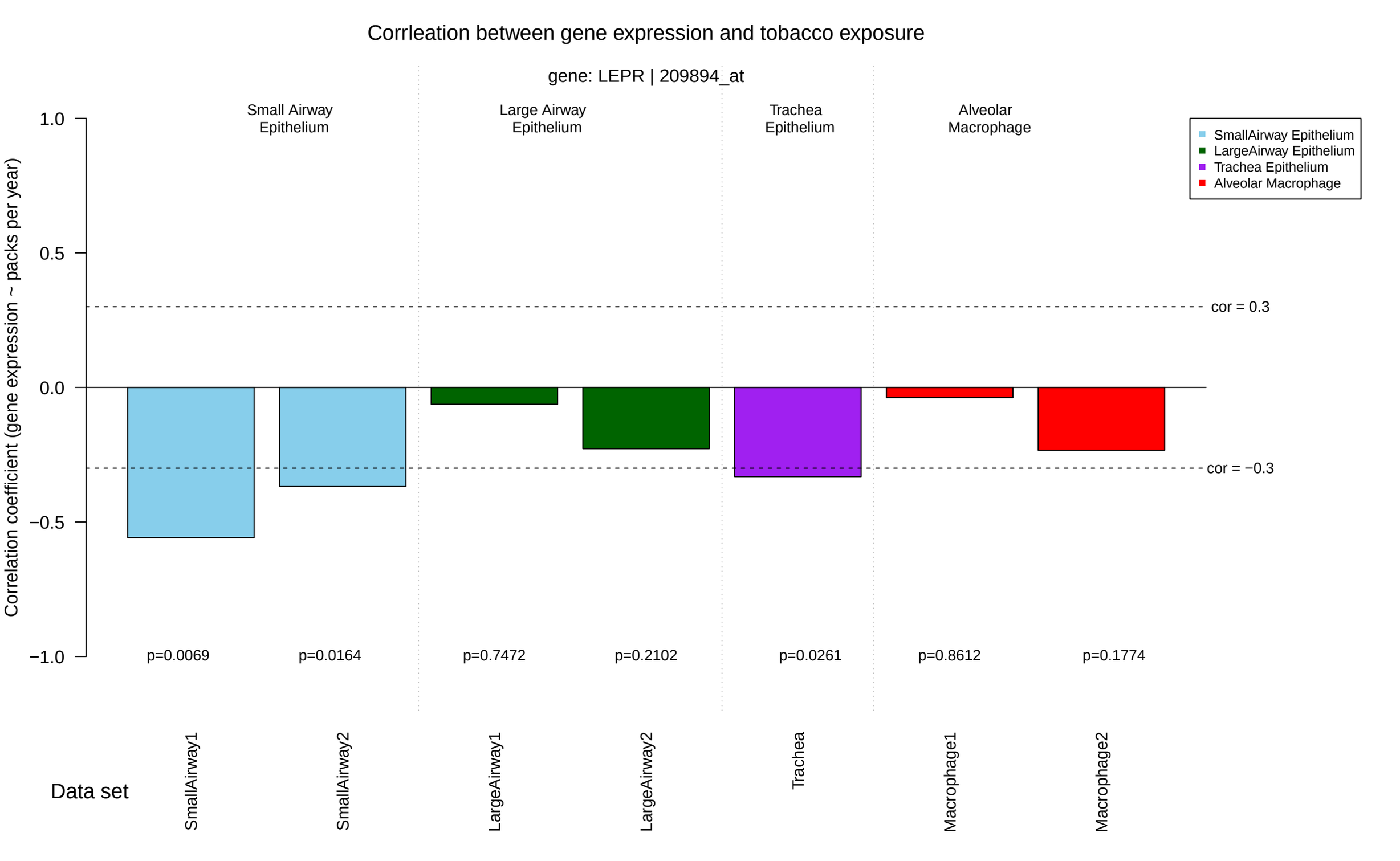

set: LargeAirway1 (p=0.0354)
LEPR | 209894_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1046)
LEPR | 209894_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0011)
LEPR | 209894_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2243)
LEPR | 209894_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2314)
LEPR | 209894_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LEPR | 209894_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0069
p=0.0164
p=0.7472
p=0.2102
p=0.0261
p=0.8612
p=0.1774
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2